ENTHUSIAST ADVANCE COURSE

PHASE : MEA & MEPS-I (A)

TARGET : PRE MEDICAL 2024

Test Type : **MINOR** Test Pattern : **NEET (UG)**

TEST DATE : 14-08-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	2	3	4	4	2	2	3	4	1	1	1	2	2	4	4	4	3	2	1	1	2	2	3	2	1	1	3	4	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	1	4	2	4	2	2	3	2	2	3	2	2	4	1	2	3	3	3	3	1	1	3	3	1	2	2	1	1	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	3	2	3	4	2	3	2	2	4	1	3	3	4	2	3	2	2	4	2	4	3	2	2	1	1	2	4	2	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	4	2	4	3	4	1	4	3	3	3	1	2	2	3	2	3	4	4	3	4	4	4	2	2	1	4	3	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	4	1	3	4	3	4	1	3	3	2	2	3	1	1	3	3	2	3	4	3	4	3	4	2	2	3	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	4	2	3	4	3	2	4	3	1	4	2	1	2	4	3	4	3	2	3	3	2	1	4	3	4	2	2	4	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	4	1	2	3	2	2	1	3	4	3	2	2	4	4	4	4	3	3	3										

HINT – SHEET

SUBJECT : PHYSICS

SECTION-A

1. **Ans (1)**

$$\lambda_{neutron} \propto \frac{1}{\sqrt{T}} \Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}}$$

$$\Rightarrow \frac{\lambda}{\lambda_2} = \sqrt{\frac{(273+927)}{(273+27)}} = \sqrt{\frac{1200}{300}} = 2 \Rightarrow \lambda_2 = \frac{\lambda}{2}$$

2. **Ans (2)**

$r \propto A^{1/3}$

3. **Ans (3)**

Nuclear force between nucleons is independent of charge so force is same for all nucleons

$F_{N-N} = F_{N-P} = F_{P-P}$

4. **Ans (4)**

$${}_1H^2 + {}_1H^2 \longrightarrow {}_2He^4 + Q$$

$$\Delta E = 4(7) - 2(1.1 + 1.1)$$

$$= 23.6 \text{ MeV}$$

5. **Ans (4)**

$${}^{A}_{Z}X \xrightarrow{\alpha-\text{decay}} {}^{A-4}_{Z-2}Y \xrightarrow{\beta^+-\text{decay}} {}^{A-4}_{Z-3}Z$$

No. of neutron $N_i = (A - Z)$, $N_F = A - 4 - (Z - 3)$

$= A - Z - 1 = N_i - 1$

6. **Ans (2)**

Here T1/2 = 20 minutes; we know $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T_{1/2}}$

For 20% decay $\frac{N}{N_0} = \frac{80}{100} = \left(\frac{1}{2}\right)^{t_1/20}$ ....(i)

For 80% decay $\frac{N}{N_0} = \frac{20}{100} = \left(\frac{1}{2}\right)^{t_2/20}$ ....(ii)

Dividing (ii) by (i)

$\frac{1}{4} = \left(\frac{1}{2}\right)^{\frac{(t_2 - t_1)}{20}}$; on solving we get $t_2 - t_1 = 40$ min.

7. Ans (2)

$A = A_0\left(\frac{1}{2}\right)^{t/TH}$

$\frac{1250}{5000} = \left(\frac{1}{2}\right)^{t/TH}$

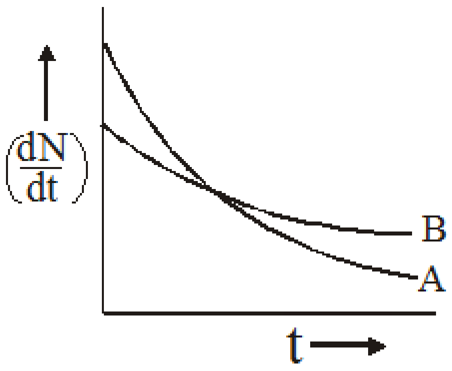
$\frac{1}{4} = \left(\frac{1}{2}\right)^{5/TH}$

$TH = \frac{5}{2} = 2.5 \text{ min}$

$\lambda = \frac{\ell n^2}{T_H} = 0.4\ \ell n2$

8. Ans (3)

From the given figure, it is clear that slope of curve A is greater than that of curve B. So rate of decay is faster for A than that of B.



We know that $\left(\frac{dN}{dt}\right) \propto \lambda$, at any instant of time hence we can say that $\lambda_A > \lambda_B$. At point P shown in the diagram the two curve intersect. Hence at point P, rate of decay for both A and B is the same.

9. Ans (4)

$\frac{dN}{dt} = -\lambda N \Rightarrow \left|\frac{dN}{dt}\right| = \frac{0.693}{T_{1/2}} \times N$

$= \frac{0.693}{1.2 \times 10^7} \times 4 \times 10^{15} = 2.3 \times 10^8 \text{ atoms / sec}$

10. Ans (1)

$\lambda_{min} = \frac{hc}{eV} = \frac{12400 V olt}{V} Å$

$\lambda_{min} = \frac{12400}{25 \times 10^3} Å = 0.49Å$

12. Ans (1)

$18 \text{ KeV} = \frac{1}{2} mV^2$

13. Ans (2)

$f = \frac{nv}{2\ell}$

$f \propto n$

14. Ans (2)

Let the frequency of standard for k = x

$n_A = \frac{102}{100}x, n_B = \frac{97}{100}x$

Number of beats per second = $n_A - n_B = 6$

$\frac{102}{100}x - \frac{97}{100}x = 6$

$x = \frac{6 \times 100}{5} = 120Hz$

$n_A = \frac{102}{100}x = 122.4Hz$

16. Ans (4)

The speed at which the wavefronts move is the speed of sound in air, which is independent of the speed of the source and the location of the observer. So the first choice is incorrect.

The observed frequency, however, is determined by the number of wavefronts passing the observer per unit time. So, the more closely the wavefronts are spaced, the higher the frequency. Inspection of the figure shows that the wavefronts are most closely spaced for observer C.

17. Ans (4)

For closed pipe (Second overtone)

$f_C = \frac{5V}{4\ell}$ ...(1)

For original open pipe first overtone

$f_0 = 2\left(\frac{V}{2\ell}\right)$ ...(2)

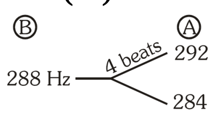
Given $f_C = 100 + f_0$

$\Rightarrow \frac{5V}{4\ell} = 100 + \frac{2V}{2\ell} \Rightarrow \frac{V}{4\ell} = 100$

Fundamental frequency of open pipe is

$\frac{V}{2\ell} = 200 \text{ Hz}$

18. Ans (3)



After loading wax, frequency of A decrease. So beats frequency will also decrease.

Ans. would be 292

19. Ans (2)

Here, $\rho = 8000\ \text{kg/m}^3$

and $B = 2 \times 10^9\ \text{N/m}^2$

Thus, $v = \sqrt{\dfrac{B}{\rho}} = \sqrt{\dfrac{2 \times 10^9 \text{N/m}^2}{8000\ \text{kg/m}^3}} = 500\ \text{m/s}$

20. Ans (1)

$$f_1 = f_0\left(\frac{320}{320-20}\right) = f_0\frac{320}{300}$$

$$f_2 = f_0\left(\frac{320}{320+20}\right) = f_0\frac{320}{340}$$

$$\frac{f_1 - f_2}{f_0} \times 100 = \left(\frac{320}{300} - \frac{320}{340}\right) \times 100$$

$= (1.066 - 0.941) \times 100 \cong 12.4\%$

21. Ans (1)

$$\text{slope} = \frac{a}{x} = -\frac{3}{4}$$

$$\Rightarrow a = \frac{-3}{4}x \Rightarrow \omega = \frac{\sqrt{3}}{2} \Rightarrow T = \frac{2\pi}{\omega}$$

22. Ans (2)

$$TE = \frac{1}{2}kA^2$$

$$7.5 = \frac{1}{2}k\frac{A^2}{4} \Rightarrow TE = 30\ J$$

23. Ans (2)

$$k_{eq} = \frac{2k_1k_2}{2k_1 + k_2}$$

24. Ans (3)

Area of sector of circle $A = \dfrac{1}{2}r^2\theta$

$A \propto r^2$

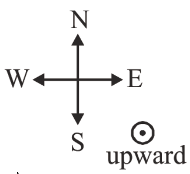
$r = \dfrac{\sqrt{2mk}}{qB}$; $r \propto \sqrt{k}$; $A \propto K$

25. Ans (2)

Lorentz force $q\vec{E} + q(\vec{V} \times \vec{B}) = \vec{0}$ remain a charge undeflected.

So $\vec{E} = -(\vec{V} \times \vec{B})$

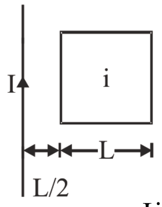
$= -(V\ \text{upward}) \times (B\ \text{east})$



$\vec{E} = VB$ south

By right hand screw rule.

27. Ans (1)



$$F_{net} = \frac{\mu_0 Ii}{2\pi\left(\frac{L}{2}\right)} - \frac{\mu_0 Ii}{2\pi\left(\frac{3L}{2}\right)} = \frac{2\mu_0 Ii}{3\pi}\ \text{Newton}$$

28. Ans (3)

Refer class notes.

SECTION-B

36. Ans (2)

$$\lambda_e = \frac{h}{\sqrt{2meV}} \quad (1)$$

$$\lambda_{pr} = \frac{h}{\sqrt{2MeV}} \quad (2)$$

eq.(2) ÷ eq.(1)

$$\frac{\lambda_{pr}}{\lambda_e} = \sqrt{\frac{m}{M}}$$

$$\Rightarrow \lambda_{pr} = \lambda\sqrt{\frac{m}{M}}$$

37. Ans (2)

$$\frac{BE}{A} = 5.6 \Rightarrow \frac{BE}{7} = 5.6 \Rightarrow BE = 39.2\ \text{MeV}$$

$$\therefore \Delta m = \frac{39.2}{931} = 0.042\ \text{amu}$$

38. Ans (3)

(1) For $1 < A < 50$, on fusion mass number for compound nucleus is less than 100

$\Rightarrow$ Binding energy per nucleon remains same

$\Rightarrow$ No energy is released

(2) For $51 < A < 100$, on fusion mass number for compound nucleus is between 100 & 200

$\Rightarrow$ Binding energy per nucleon increases

$\Rightarrow$ Energy is released.

(3) For $100 < A < 200$, on fission, the mass number of product nuclei will be between 50 & 100

$\Rightarrow$ Binding energy per nucleon decreases

$\Rightarrow$ No energy is released

(4) For $200 < A < 260$, on fission, the mass number of product nuclei will be between 100 & 130

$\Rightarrow$ Binding energy per nucleon increases

$\Rightarrow$ Energy is released.

39. Ans (2)

$P = \frac{nE}{t} \Rightarrow 300 \times 10^6 = \frac{n \times 170 \times 10^6 \times 1.6 \times 10^{-19}}{t}$

$\therefore$ Number of atoms per sec $\frac{n}{t} = 1.102 \times 10^{19}$

Number of atoms per hour $= 1.02 \times 10^{19} \times 3600$

$= 3.97 \times 10^{22}$.

40. Ans (2)

Recoil energy $= \frac{p^2}{2M}$

$p = \frac{h}{\lambda} = \frac{h\nu}{c}$

$\Rightarrow$ Recoil energy $= \frac{h^2\nu^2}{2Mc^2}$

41. Ans (3)

$L = 10 \log_{10}\left(\frac{I}{I_0}\right)$

$\therefore \quad L = 10 \log_{10}\left(\frac{2 \times 10^{-8}}{10^{-12}}\right)$

$(\because I_0 = 10^{-12}\ W/m^2)$

$L = 10 \log_{10}(2 \times 10^4)$

$L = 10 [\log_{10}2 + \log_{10}(10)^4]$

$L = 10\ (4.3)$

$L = 43$ decibel

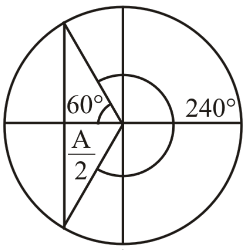
42. Ans (2)

$PE_{avg} = \frac{1}{4}m\omega^2A^2$

$= \frac{1}{4}m \times 4\pi^2f^2A^2$

$= m\pi^2f^2A^2$

43. Ans (2)



$T' = \frac{240°}{360°} \times T = \frac{3}{2} \times 2a\sqrt{\frac{1}{a^2}}$

$T' = \frac{4}{3}$ sec

TG: @Chalnaayaaar

44. Ans (4)

Since $A_t = A_0 e^{\frac{-bt}{2m}}$

When t = 2 sec, $\frac{A}{3} = Ae^{\frac{-2b}{2m}}$

$\frac{1}{3} = e^{-b/m}$

When t = 6 sec

$\frac{A_0}{n} = A_0 e^{\frac{-6b}{2m}}$

$\frac{1}{n} = \left(e^{\frac{-b}{m}}\right)^3$

$\frac{1}{n} = \left(\frac{1}{3}\right)^3$

$n = 3^3$.

SUBJECT : CHEMISTRY

SECTION-A

52. Ans (1)

$\left(\frac{w}{E}\right)_{Cl} = \left(\frac{w}{E}\right)_{\text{metal chloride}}$

$\frac{8.35}{35.5} = \left(\frac{10}{E_M + 35.5}\right)$

$E_M = 7g$

$x = \frac{2 \times 85}{7 + 35.5} = 4 \longrightarrow MCl_4$

54. Ans (3)

$CaCO_3\ (S) \xrightarrow{\Delta} CO_2(g) + CaO(s)$

100 kg $\longrightarrow$ 44 kg

$\frac{80}{100} \times 10kg \rightarrow \frac{44}{100} \times \frac{80}{100} \times 10 = 3.52$ kg

55. Ans (1)

Number of electron is one ion of $CO_3^{-2} = 6 + 24 + 2$

$= 32$

Number of ions in 6 mg of $CO_3^{-2} = \frac{6 \times 10^{-3}}{60} \times 6 \times 10^{23}$

$= 6 \times 10^{19}$

$\therefore$ No. of electrons in 6 mg $CO_3^{-2} = 6 \times 10^{19} \times 32$

$= 192 \times 10^{19}$

$= 1.92 \times 10^{21}$

59. Ans (1)

As, TF $\propto \frac{1}{n}$

$Al_2(SO_4)_3$ will give the maximum number of particles on dissociation, hence it gives lowest freezing point.

60. Ans (2)

$$M_{H_2O_2} = \frac{V.S}{11.2} = \frac{10.08}{11.2} = 0.9\,M$$

$$M = \frac{\left(\frac{wt}{34}\right)}{250} \times 1000 = 0.9$$

wt = 7.65 gm

62. Ans (3)

NCERT-XII, Pg. # 43

$$Y_A = \frac{X_A P_A^o}{X_A P_A^o + X_B P_B^o}$$

$$= \frac{1}{1 + \frac{X_B P_B^o}{X_A P_A^o}} = \frac{1}{1 + \frac{2}{1} \times \frac{2}{1}} = \frac{1}{5} = 0.2$$

63. Ans (2)

NCERT-XII Pg#194

$$N_2(g) + 3H_2(g) \leftrightharpoons 2NH_3(g)$$

$$K_C = \frac{[NH_3]^2}{[N_2]\,[H_2]^3} = \frac{(2 \times 10^{-2})^2}{(2 \times 10^{-2})\,(4 \times 10^{-2})^3}$$

$$\Rightarrow \frac{2 \times 10^{-2}}{64 \times 10^{-6}}$$

$\Rightarrow 0.0312 \times 10^4$

$\Rightarrow 312$

64. Ans (3)

NCERT-XII Pg#196

$K_p = K_c\,(RT)^{\Delta ng}$ $[\Delta ng = 3 - 2 = 1]$

$$K_p = (3 \times 10^{-4}) \left[\frac{1}{12} \times 1000\right]^1$$

$$= \frac{1}{4} \times 10^{-1} = 0.025$$

65. Ans (4)

NCERT-XII Pg-203,204

66. Ans (2)

NCERT-XII, Pg. # 58

$\Delta T_f = i\,k_f \times m$

$7.1 \times 10^{-3} = i \times 1.86 \times 0.001$

$i = 3.817$

$i = 1 + (n - 1) \propto$

$i = n = 3.817 + \simeq 4$

$3K^+ + [Fe(CN)_6]^{-3}$

67. Ans (3)

NCERT-XII Pg#196

$$SO_2 + NO_2 \rightleftharpoons SO_3 + NO$$

	SO_2	NO_2	SO_3	NO
t = 0	1	1	1	1
at eq.	1–x	1–x	1+x	1+x

$$K_c = \frac{[SO_3]\,[NO]}{[SO_2]\,[NO_2]} = \frac{(1+x)(1+x)}{(1-x)(1-x)}$$

$$16 = \frac{(1+x)^2}{(1-x)^2}$$

$$4 = \frac{1+x}{1-x}$$

$4.4x = 1 + x$

$3 = 5x$

$$x = \frac{3}{5} = 0.6$$

$(NO) = 1 + x = 1 + 0.6 = 1.6$

$(NO_2) = 1 - x = 1 - 0.6 = 0.4$

68. Ans (2)

NCERT-XII Pg#202,203

69. Ans (2)

NCERT-XII Pg# 202

70. Ans (4)

NCERT-XII Pg#45

71. Ans (1)

NCERT-XII Pg#35

$$m = \frac{1000 \times x_B}{(1 - x_B) \times (M_w)_A}$$

$$= \frac{1000 \times 0.2}{0.8 \times 78} = 3.2$$

72. Ans (3)

NCERT-XII Pg#46

76. Ans (3)

$$\text{Stability of carbocation} \propto \frac{+M, +H, +I}{-M, -I}$$

SECTION-B

88. Ans (4)

A	+	3B	$\rightleftharpoons$	2C	+	D
2b		b		0		0
2b–x		b–3x		2x		x

$b - 3x = 2x$

$b = 5x$

$$\%\,B(\text{reacted}) = \frac{3x}{b} \times 100$$

$$= \frac{3x}{5x} \times 100 = 60$$

90. **Ans (1)**
NCERT-XII Pg#201
$\Delta G^0 = -2.303 \text{ RT} \log K_{eq}$
$= -2.303 \times 8.314 \times 500 \log(2 \times 10^{10})$
$= -2.303 \times 8.314 \times 500 [\log 2 + 10]$
$= -2.303 \times 8.314 \times 500 \times 10.30$
$= -98.6$ KJ/mol

91. **Ans (1)**
NCERT-XII Pg#193

92. **Ans (3)**
NCERT-XII Pg#56

93. **Ans (4)**
NCERT-XII Pg#54

94. **Ans (2)**
NCERT-XII Pg#56

SUBJECT : BIOLOGY-I

SECTION-A

101. **Ans (3)**
NCERT XI, PAGE NO. 96

102. **Ans (3)**
NCERT XI, PAGE NO. 96

103. **Ans (1)**
NCERT - XI, PAGE NO. 87

104. **Ans (2)**
NCERT XI, Pg # 97

105. **Ans (2)**
NCERT XI, Pg. # 93, 94

106. **Ans (3)**
NCERT XI, Pg. # 85

107. **Ans (2)**
Module–3, Pg # 148, 149, 150

128. **Ans (4)**
NCERT Page No. 167

129. **Ans (1)**
NCERT Pg. # 169

130. **Ans (3)**
NCERT-XII Pg. # 169/183(H) Para:9.1.2

131. **Ans (3)**
NCERT Pg. # 167, 168

132. **Ans (2)**
NCERT Pg#169,167

133. **Ans (2)**
NCERT Pg#169

SECTION-B

136. **Ans (1)**
NCERT XI PAGE NO. 91
Functions of pericycle in a dicot root
(i) Formation of lateral roots
(ii) Formation of a part of vasculer cambium
(iii) Formation of cork cambium

137. **Ans (3)**
NCERT - XI, PAGE NO. 86, 87

138. **Ans (3)**
NCERT XI, Pg. # 97

139. **Ans (2)**
NCERT XII Pg.# 138,139

140. **Ans (3)**
NCERT(XII) Pg#127 Para:7.1

141. **Ans (4)**
NCERT(XII) Pg# 131 Fig: 7.3

146. **Ans (2)**
NCERT Page 132, 4th line, Last Para

148. **Ans (3)**
NCERT PAGE : 169

149. **Ans (2)**
NCERT-Pg#167

150. **Ans (4)**
NCERT Pg#168

SUBJECT : BIOLOGY-II

SECTION-A

151. **Ans (4)**
NCERT-XII Pg # 199

157. **Ans (2)**
NCERT XII Pg. # 174

159. **Ans (3)**
NCERT XI Pg.# 148

162. **Ans (2)**
NCERT–XII, Pg. # 175

163. **Ans (1)**
NCERT–XII, Pg. # 182,183

167. **Ans (4)**
NCERT XII Pg # 198

171. **Ans (3)**
NCERT (XIIth) Pg. # 196,199

173. **Ans (1)**
NCERT (XIth) Pg. # 232

174. **Ans (4)**
NCERT-XI Pg. # 233

175. **Ans (3)**
NCERT (XIth) Pg. # 233

176. **Ans (4)**
NCERT-XI, Pg. # 229, 234

177. **Ans (2)**
NCERT Page # 178

178. **Ans (2)**
NCERT-XI, Pg. # 189

180. **Ans (4)**
NCERT-XI, Pg. # 191

181. **Ans (2)**
NCERT-XI, Pg. # 180

183. **Ans (1)**
NCERT XI Pg. # 230

184. **Ans (2)**
NCERT Pg. # 232

185. **Ans (3)**
NCERT-XI, Pg. # 189

SECTION-B

187. **Ans (2)**
NCERT (XIIth) Pg. # 198

193. **Ans (2)**
NCERT (XII) Pg. # 173

196. **Ans (4)**
NCERT-XI, Pg. # 237

198. **Ans (3)**
NCERT-XI, Pg. # 192

199. **Ans (3)**
NCERT-XI, Pg. # 235

200. **Ans (3)**
NCERT-XI, Pg.# 233